

Project 3 - Host a Static Website Using Amazon S3 Report

Introduction

Cloud computing enables organizations and individuals to deploy applications and websites without maintaining physical servers. Amazon S3 (Simple Storage Service) is a highly scalable object storage service provided by AWS that can also be used to host static websites. Static website hosting allows users to serve HTML, CSS, JavaScript, images, and other static content directly from an S3 bucket through a public URL. This project focuses on configuring an S3 bucket to host a static website and make it accessible over the internet.

Objective

The main objective of this project is to:

1. Create an Amazon S3 bucket
2. Upload website files (HTML/CSS)
3. Enable Static Website Hosting
4. Configure bucket permissions for public access
5. Access the website using the generated S3 website endpoint
6. Understand static website deployment using cloud services

Overview of Amazon S3 Static Website Hosting

Amazon S3 provides a feature called Static Website Hosting, which enables users to host web pages directly from an S3 bucket without requiring a web server.

Key Concepts

- ✧ **Bucket:** A container used to store website files and objects
- ✧ **Object:** Individual files such as HTML, CSS, JavaScript, and images
- ✧ **Static Website Hosting:** A feature that allows S3 buckets to serve web content through a public URL
- ✧ **Bucket Policy:** A JSON-based access control mechanism used to grant public access
- ✧ **Website Endpoint:** A URL generated by AWS for accessing the hosted website

Methodology

Bucket Creation

A new Amazon S3 bucket was created using the AWS Management Console. A globally unique bucket name was assigned, and the Asia Pacific (Sydney) region was selected. Public access settings were configured to allow website hosting.

Uploading Website Files

An HTML file named `index.html` was created and uploaded to the S3 bucket. This file serves as the main webpage of the hosted website.

Enabling Static Website Hosting

The Static Website Hosting feature was enabled from the bucket properties section. The uploaded `index.html` file was configured as the index document for the website.

Configuring Public Access

A bucket policy was added to grant public read access using the `s3:GetObject` permission. This allowed users to access website files through a web browser.

Accessing the Website

The generated S3 website endpoint URL was opened in a browser. The website loaded successfully, confirming that the hosting configuration was working correctly.

Results

1. Successfully created an Amazon S3 bucket
2. Uploaded website files to the bucket
3. Enabled Static Website Hosting
4. Configured bucket policies for public access
5. Generated and accessed the S3 website endpoint URL
6. Successfully hosted a publicly accessible static website

Key Learnings

- ✧ Amazon S3 can be used to host static websites without dedicated web servers
- ✧ Static Website Hosting must be enabled explicitly in bucket properties
- ✧ Public access requires proper bucket policy configuration
- ✧ Bucket policies control access permissions to objects stored in S3
- ✧ S3 website endpoints provide direct access to hosted web content

Advantages of Amazon S3 Static Website Hosting

- ✧ Easy to configure and deploy
- ✧ Highly scalable and durable
- ✧ Cost-effective hosting solution
- ✧ No server management required
- ✧ Integrates seamlessly with other AWS services

Limitations

- ✧ Supports only static content
- ✧ Does not support server-side processing
- ✧ Requires proper permission configuration for public access
- ✧ Website endpoints use HTTP by default unless integrated with additional AWS services
- ✧ Advanced features may require CloudFront and Route 53 integration

Conclusion

This project provided practical experience in hosting a static website using Amazon S3. It demonstrated how cloud storage services can be utilized for web hosting, how bucket policies control access permissions, and how AWS enables scalable and cost-effective website deployment. The project enhanced understanding of cloud-based web hosting and AWS storage services.